

- RAG Learning

Homeostasis and Response

Edexcel A A-Level Biology — Topic 7

7.1 Principles of Homeostasis

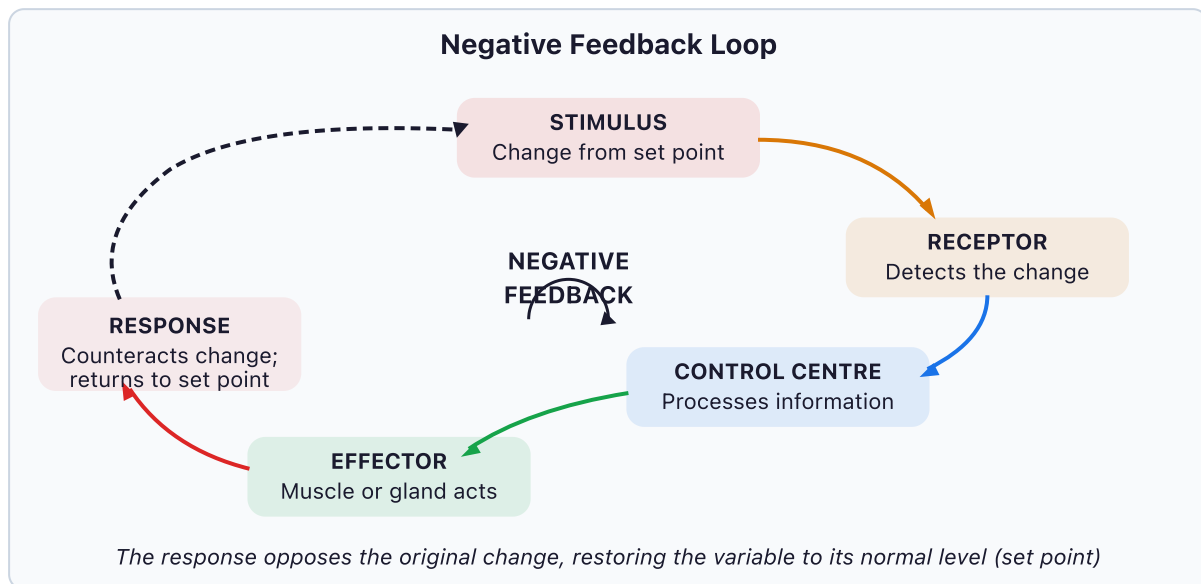
Homeostasis: The maintenance of a constant internal environment within restricted limits, despite changes in the external environment. The internal environment includes blood glucose concentration, body temperature, blood pH, water potential, and CO₂ concentration. Homeostasis is essential because enzyme-controlled reactions require specific conditions (temperature, pH) to function at optimal rates. Significant deviations denature enzymes, alter membrane fluidity, or change water potential — all potentially fatal.

Negative Feedback

Negative feedback is the fundamental mechanism of homeostasis. It detects a deviation from the **set point** (normal value) and activates responses that *oppose* the change, returning the variable to its optimal level. The system operates as a continuous loop:

1. **Stimulus** — a change in the internal environment away from the set point (e.g. blood glucose rises after a meal)
2. **Receptor** — a sensory cell or structure that detects the change (e.g. beta cells in pancreatic islets detect elevated glucose)
3. **Coordination centre** — processes the information and determines the appropriate response (e.g. hypothalamus for temperature, pancreas for glucose, medulla oblongata for heart rate)
4. **Effector** — carries out the corrective response; either a muscle (contracts) or a gland (secretes a hormone)
5. **Response** — counteracts the original change, restoring the variable towards the set point

Because negative feedback takes time to act and then correct, homeostatic variables **oscillate** around the set point rather than being held at a fixed value. The system is self-correcting: as the variable returns to normal, the stimulus diminishes and the response decreases.



Positive Feedback

In positive feedback, the response **amplifies** the original change, moving the variable further from the set point. This creates an escalating cascade that continues until an external event breaks the loop. Positive feedback is rare in homeostasis but critical in specific biological events:

- **Oxytocin during childbirth:** Pressure of the baby's head on the cervix stimulates oxytocin release from the posterior pituitary → stronger uterine contractions → more pressure → more oxytocin. The loop breaks when the baby is delivered
- **Blood clotting cascade:** Platelet activation releases chemicals that activate more platelets, amplifying the clot formation at the wound site
- **Depolarisation during an action potential:** Opening of Na^+ channels causes depolarisation, which opens more Na^+ channels (positive feedback during the rising phase). The loop ends when Na^+ channels inactivate
- **Fruit ripening:** Ethylene released by ripening fruit stimulates more ethylene production in nearby fruit

Exam Tip: Positive feedback is not a failure of homeostasis — it serves specific functions where a rapid, amplified response is needed. Always contrast it with negative feedback:

negative = stability (opposes change), positive = completion of a process (amplifies change until an endpoint is reached).

7.2 The Nervous System

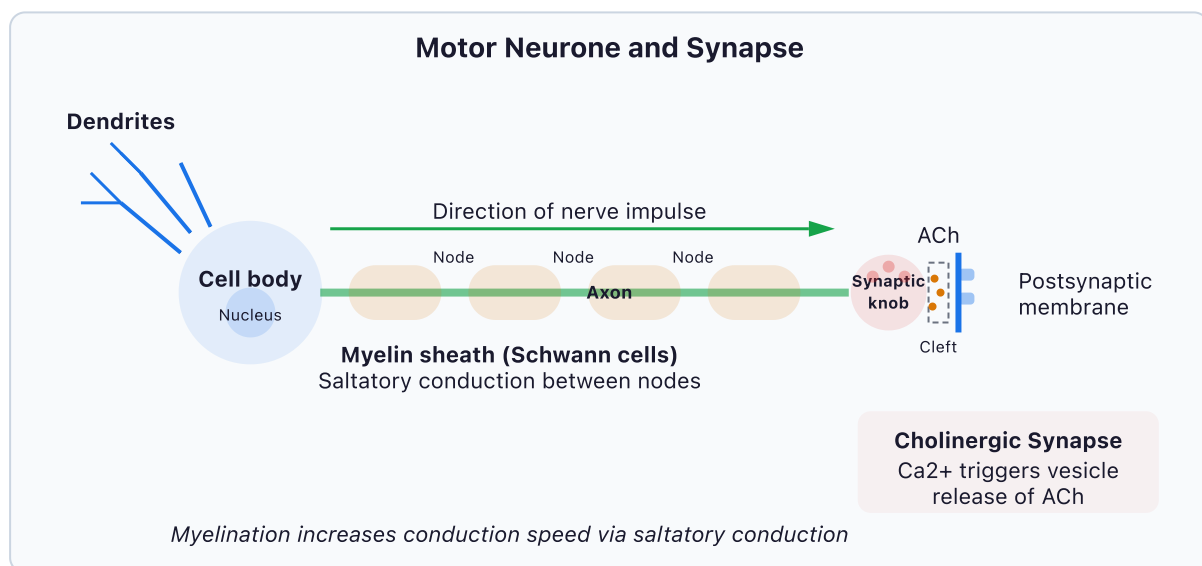
Nervous system: A rapid, short-lived communication system using electrical impulses transmitted along neurones. It enables rapid coordination of responses to stimuli. The nervous system is divided into: **CNS** (central nervous system — brain and spinal cord, processes information) and **PNS** (peripheral nervous system — sensory and motor neurones connecting receptors and effectors to the CNS). The PNS is further divided into the **somatic** (voluntary, skeletal muscle) and **autonomic** (involuntary, smooth muscle/glands/cardiac) nervous systems. The autonomic system comprises the **sympathetic** (fight-or-flight) and **parasympathetic** (rest-and-digest) divisions.

Neurone Structure

Type	Function	Key Structural Features	Location
Sensory neurone	Transmits impulses from receptor to CNS	Long dendron carrying impulse from receptor; cell body located off to one side (in dorsal root ganglion); short axon entering spinal cord	PNS → CNS
Motor neurone	Transmits impulses from CNS to effector (muscle/gland)	Many short dendrites; cell body in CNS (ventral horn); long myelinated axon terminating at motor end plate or gland	CNS → PNS
Relay (interneurone)	Connects sensory and motor neurones within CNS; integrates information	Short axon; many connections (synapses); found entirely within CNS	Within CNS only

Key Neurone Features

- **Dendrites/dendrons:** Carry impulses towards the cell body; have many branches to receive signals from multiple sources
- **Axon:** Carries impulses away from cell body; can be up to 1 m long (e.g. sciatic nerve from spinal cord to foot)
- **Myelin sheath:** Formed by **Schwann cells** wrapping around the axon in concentric layers of lipid-rich membrane. Acts as electrical insulator, forcing the impulse to "jump" between gaps (nodes of Ranvier) — **saltatory conduction**. Increases speed from ~2 m/s (unmyelinated) to ~120 m/s (myelinated)
- **Nodes of Ranvier:** Gaps between Schwann cells (~1 mm apart) where the axon membrane is exposed and contains a high density of voltage-gated Na^+ channels
- **Synaptic knob (terminal bouton):** Swollen end of axon containing mitochondria (ATP for vesicle transport), synaptic vesicles (containing neurotransmitter), and voltage-gated Ca^{2+} channels



Resting Potential (–70 mV)

At rest, the neurone membrane is **polarised** at approximately –70 mV (inside negative relative to outside). This potential difference is established and maintained by:

- **Na^+/K^+ ATPase pump:** Actively transports 3 Na^+ out for every 2 K^+ in (electrogenic — net export of positive charge). Uses ~40% of neural ATP

- **Potassium leak channels:** The membrane is ~50x more permeable to K^+ than Na^+ at rest, so K^+ continually leaks out down its concentration gradient, carrying positive charge out of the cell
- **Large organic anions (A^-):** Proteins and other negatively charged molecules trapped inside the cell cannot cross the membrane, contributing to the negative internal charge
- **Na^+ channels are mostly closed at rest:** Voltage-gated Na^+ channels remain shut, preventing Na^+ influx that would depolarise the membrane

Action Potential

An action potential is a rapid, transient reversal of the membrane potential that propagates along the axon as a nerve impulse. It follows an **all-or-nothing** principle — if the threshold is reached, the full action potential fires; if not, nothing happens. The strength of stimulus is coded by the **frequency** of action potentials, not their amplitude.

1. **Resting state:** Membrane at -70 mV; Na^+ channels closed; K^+ leak channels open
2. **Depolarisation:** A stimulus (e.g. neurotransmitter binding, pressure) causes some Na^+ channels to open; Na^+ floods in down its electrochemical gradient; membrane becomes less negative (depolarises)
3. **Threshold (-55 mV):** If depolarisation reaches the threshold, voltage-gated Na^+ channels open rapidly (**positive feedback**); massive Na^+ influx drives membrane potential to approximately $+40$ mV
4. **Repolarisation:** After ~ 1 ms, Na^+ channels **inactivate** (distinct from closing — a separate inactivation gate blocks the channel); voltage-gated K^+ channels open (with a slight delay); K^+ flows out down its concentration gradient; membrane repolarises
5. **Hyperpolarisation (undershoot):** K^+ channels are slow to close, so excess K^+ leaves; membrane briefly becomes more negative than -70 mV (typically -80 to -90 mV)
6. **Resting potential restored:** K^+ channels close; Na^+/K^+ ATPase pump re-establishes ion gradients; Na^+ channel inactivation gates reset



Refractory Period

The period following an action potential during which the neurone cannot (or is less likely to) fire again:

- **Absolute refractory period (~1 ms):** Na⁺ channels are inactivated and cannot reopen regardless of stimulus strength. No action potential is possible
- **Relative refractory period (~2–4 ms):** Some Na⁺ channels have reset but K⁺ channels are still open (hyperpolarisation). A stronger-than-normal stimulus is needed to reach threshold

The refractory period is functionally important because it: (1) ensures impulses travel in **one direction only** (the region just depolarised is refractory), (2) ensures action potentials are **discrete, separate events** that cannot merge, (3) limits the **maximum frequency** of impulses (typically 500–1000 per second).

Factors Affecting Transmission Speed

Factor	Effect	Explanation
Myelination	Much faster (up to 120 m/s vs 2 m/s)	Saltatory conduction: impulse jumps between nodes of Ranvier, bypassing myelinated sections. Ion exchange only occurs at nodes, requiring fewer ions to be moved

Axon diameter	Larger = faster	Greater cross-sectional area means less electrical resistance to ion flow along the axon interior (like a wider pipe carrying more water)
Temperature	Higher = faster (up to ~40 degrees C)	Increased kinetic energy: ions diffuse faster, enzymes (Na/K ATPase) work faster, channel proteins change conformation faster. Above ~40 degrees C, proteins denature

7.3 Synapses

Synapse: The junction between two neurones (or a neurone and an effector cell) across which a nerve impulse is transmitted. Most synapses are **chemical synapses** using neurotransmitter molecules that diffuse across a ~20 nm gap called the **synaptic cleft**. The most common excitatory neurotransmitter in the PNS is **acetylcholine (ACh)**; synapses using ACh are called **cholinergic synapses**.

Cholinergic Synapse Transmission (Step by Step)

1. **Action potential arrives** at the presynaptic knob (synaptic terminal)
2. Depolarisation of the presynaptic membrane opens **voltage-gated Ca^{2+} channels**; Ca^{2+} floods into the synaptic knob down its concentration gradient
3. The influx of Ca^{2+} causes **synaptic vesicles** (each containing ~10,000 ACh molecules) to move towards and **fuse** with the presynaptic membrane (exocytosis). This process requires the SNARE protein complex
4. **ACh is released** into the synaptic cleft by exocytosis
5. ACh molecules **diffuse** across the ~20 nm cleft (takes <0.5 ms) and bind to **nicotinic ACh receptors** (ligand-gated ion channels) on the postsynaptic membrane. Binding is complementary and specific
6. The receptors are **Na^+ channels** that open when ACh binds; Na^+ influx causes **depolarisation** of the postsynaptic membrane (an excitatory postsynaptic potential, EPSP)

7. If the EPSP reaches **threshold**, a new action potential is generated in the postsynaptic neurone
8. **Acetylcholinesterase (AChE)** in the synaptic cleft rapidly **hydrolyses ACh** into choline + ethanoic acid (acetyl). This stops continuous stimulation. Choline is actively reabsorbed into the presynaptic knob and recycled to make new ACh (using acetyl-CoA from mitochondria)

Exam Tip: Key reasons why synaptic transmission is important: (1) ensures **unidirectional** transmission (receptors only on postsynaptic membrane, vesicles only in presynaptic knob), (2) allows **integration** of signals (summation), (3) allows **amplification** (one vesicle releases many neurotransmitter molecules), (4) provides a point for **modulation** by drugs, (5) allows **divergence** (one neurone connects to many) and **convergence** (many neurones connect to one).

Summation

A single synaptic vesicle may produce an EPSP too small to reach threshold alone. Summation is the process by which multiple small EPSPs combine to reach threshold:

- **Temporal summation:** Multiple action potentials arrive in rapid succession from the *same* presynaptic neurone, each releasing more neurotransmitter. The EPSPs add up before the previous one has decayed completely. Analogous to repeatedly tapping a drum faster and faster until the vibration builds up
- **Spatial summation:** Multiple *different* presynaptic neurones release neurotransmitter simultaneously onto the same postsynaptic neurone. The combined EPSPs from different synapses sum together to reach threshold

Inhibitory synapses also exist: they release inhibitory neurotransmitters (e.g. GABA, glycine) that open Cl^- channels or K^+ channels, causing **hyperpolarisation** (inhibitory postsynaptic potential, IPSP). The postsynaptic neurone integrates all EPSPs and IPSPs to determine whether to fire.

Drugs and Synapses

Drug Action	Mechanism	Example	Clinical/Real-World Use
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Agonist	Mimics neurotransmitter shape; binds to and activates receptor	Nicotine (mimics ACh at nicotinic receptors)	Nicotine causes dopamine release in reward pathway → addiction
Antagonist	Binds to receptor without activating it; competitively blocks neurotransmitter	Curare, atropine	Curare blocks ACh receptors at NMJ → muscle paralysis (used on poison darts); atropine dilates pupils
Inhibits breakdown	Blocks the enzyme that degrades neurotransmitter in the cleft	Nerve agents (sarin) — inhibit AChE; donepezil	Nerve agents cause continuous stimulation → paralysis/death; donepezil used in Alzheimer's treatment
Inhibits reuptake	Blocks transporter proteins that reabsorb neurotransmitter into presynaptic neurone	SSRIs (fluoxetine/Prozac); cocaine	SSRIs increase serotonin in cleft → treat depression; cocaine blocks dopamine reuptake → euphoria
Stimulates release	Causes increased vesicle fusion and neurotransmitter release	Amphetamines; black widow spider venom	Amphetamines increase dopamine/noradrenaline release
Blocks vesicle release	Prevents vesicle fusion with presynaptic membrane	Botulinum toxin (Botox)	Prevents ACh release at NMJ → muscle paralysis; used cosmetically and to treat muscle spasms

7.4 Hormonal Coordination

Endocrine vs Nervous System

Feature	Nervous System	Endocrine System
Communication method	Electrical impulses along neurones	Chemical hormones secreted into blood
Speed of signal	Very fast (milliseconds)	Slower (seconds to hours to reach target)
Duration of effect	Short-lived (stops when impulse stops)	Longer-lasting (hours, days, or permanent)
Target specificity	Precise (specific effector via direct neurone connection)	Widespread (all cells with the specific receptor are affected)
Nature of signal	Electrical (action potentials) + chemical (at synapses)	Chemical only (hormones)
Pathway	Along neurones (fixed pathways)	Via blood (reaches all parts of body)

In reality, the two systems work together. For example, the **hypothalamus** is a neuroendocrine organ that receives nervous input and produces hormones, linking the nervous and endocrine systems. The **adrenal medulla** releases adrenaline in response to sympathetic nerve stimulation.

Hormones: Peptide vs Steroid

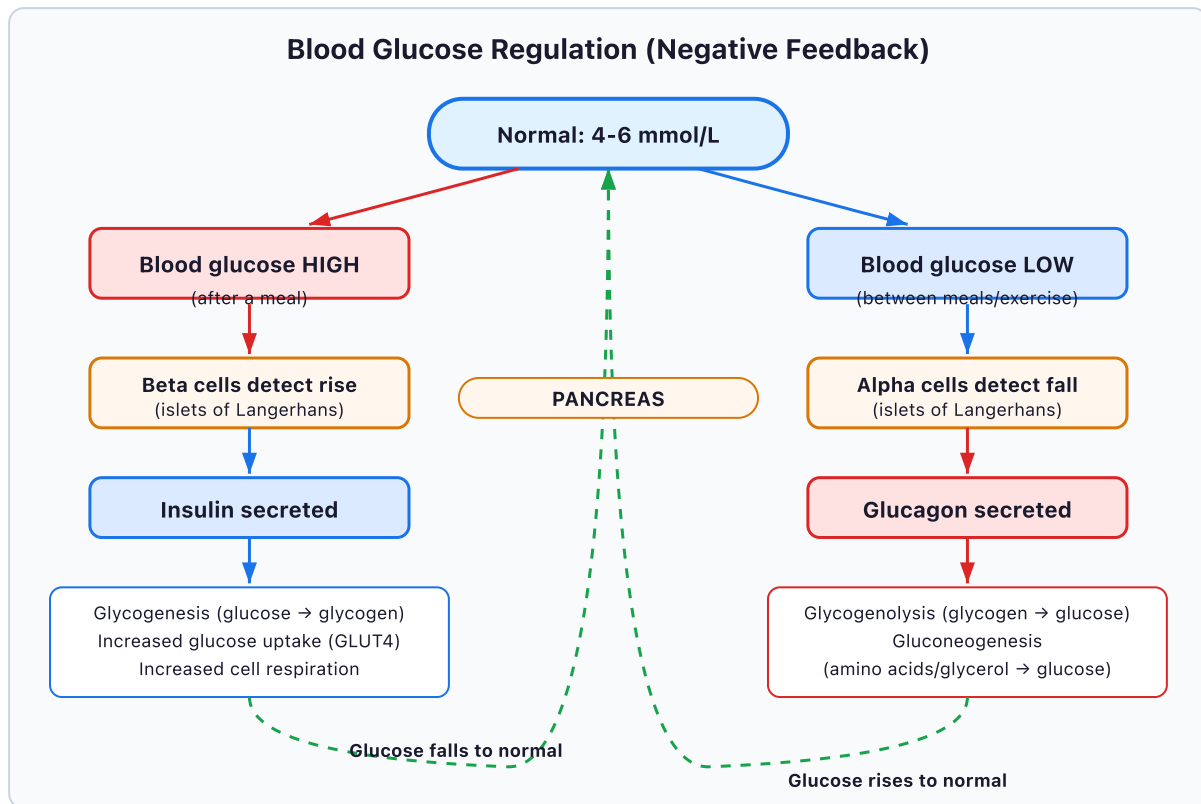
Feature	Peptide/Protein Hormones	Steroid Hormones
Chemical nature	Chains of amino acids (proteins or short peptides)	Derived from cholesterol (lipids)
Solubility	Water-soluble; insoluble in lipids	Lipid-soluble; insoluble in water
Transport in blood	Dissolved freely in plasma	Bound to carrier proteins (e.g. SHBG)

Receptor location	Cell-surface receptors (cannot cross membrane)	Intracellular receptors (cross membrane by diffusion)
Mechanism	Bind to receptor → activate G-protein → activate adenylyl cyclase → cAMP (secondary messenger) → activates protein kinases → cellular response	Cross membrane → bind intracellular receptor → hormone-receptor complex enters nucleus → acts as transcription factor → activates/represses gene expression
Speed of response	Fast (seconds to minutes)	Slow (hours to days, involves gene expression)
Examples	Insulin, glucagon, ADH, adrenaline	Oestrogen, testosterone, cortisol, progesterone

Exam Tip: A common exam question involves the secondary messenger model for peptide hormones. Remember the sequence: hormone binds receptor → G-protein activates → adenylyl cyclase converts ATP to cAMP → cAMP activates protein kinases → cascade amplification. One hormone molecule can trigger production of millions of product molecules through this cascade. This is called **signal amplification**.

7.5 Blood Glucose Regulation

Blood glucose homeostasis: The maintenance of blood glucose concentration at approximately 4–6 mmol/L (fasting) or up to ~8 mmol/L shortly after a meal. This is primarily controlled by the antagonistic hormones **insulin** (lowers glucose) and **glucagon** (raises glucose), both secreted by the **islets of Langerhans** in the pancreas. Glucose is the primary respiratory substrate for cells (especially the brain, which cannot easily use alternatives).



After a Meal (High Blood Glucose)

- Beta cells** (beta-islets) in the islets of Langerhans detect high blood glucose. The mechanism: glucose enters beta cells via GLUT2 transporters → increased glycolysis and ATP production → ATP-sensitive K^+ channels close → cell depolarises → voltage-gated Ca^{2+} channels open → Ca^{2+} triggers exocytosis of insulin-containing vesicles
- Beta cells secrete **insulin** (a peptide hormone of 51 amino acids) into the blood
- Insulin binds to **tyrosine kinase receptors** on the surface of target cells (liver, muscle, adipose tissue)
- Effects of insulin:
 - Stimulates insertion of **GLUT4 glucose transporters** into cell membranes of muscle and adipose cells, increasing glucose uptake
 - Activates enzymes for **glycogenesis** (glucose → glycogen) in liver and muscle cells
 - Increases rate of **cellular respiration** (glucose used as fuel)
 - Promotes conversion of glucose to **fatty acids** (lipogenesis) in adipose tissue
 - Stimulates amino acid uptake and protein synthesis
- Blood glucose concentration falls back to normal → less insulin secreted (negative feedback)

Between Meals / During Exercise (Low Blood Glucose)

1. **Alpha cells** (alpha-islets) detect low blood glucose
2. Alpha cells secrete **glucagon** (a peptide hormone of 29 amino acids) into the blood
3. Glucagon binds to **G-protein coupled receptors** on liver cell (hepatocyte) membranes, activating the cAMP secondary messenger cascade
4. Effects of glucagon (primarily on liver):
 - **Glycogenolysis**: glycogen → glucose (glycogen phosphorylase activated)
 - **Gluconeogenesis**: amino acids, glycerol, and lactate → glucose
 - Decreased glycogenesis (opposes insulin's effect)
5. Glucose released from hepatocytes into the blood → blood glucose rises to normal → less glucagon secreted (negative feedback)

Adrenaline also raises blood glucose during stress/exercise (fight-or-flight response) by stimulating glycogenolysis and inhibiting insulin secretion. This is a complementary system to glucagon.

Exam Tip: Remember the distinction: **glycogenesis** (glucose → glycogen, stimulated by insulin), **glycogenolysis** (glycogen → glucose, stimulated by glucagon), **gluconeogenesis** (non-carbohydrate sources → glucose, stimulated by glucagon). These are commonly confused in exams. Also note that glucagon acts primarily on the **liver** (hepatocytes), not muscle cells. Muscle glycogen is used locally and cannot release glucose into the blood.

Diabetes Mellitus

Feature	Type 1 Diabetes	Type 2 Diabetes
Cause	Autoimmune destruction of beta cells by T cells; little or no insulin produced	Target cells become resistant to insulin (insulin receptors downregulated or less responsive); beta cell function may decline over time
Onset	Usually childhood/adolescence; rapid onset	Usually adults >40 (increasingly younger); gradual onset over years

Risk factors	Genetic predisposition (HLA genes); possible viral trigger	Obesity (especially visceral fat), sedentary lifestyle, genetic predisposition, age, ethnicity
Insulin production	None or negligible	Produced but cells do not respond adequately; may eventually decline
Treatment	Insulin injections (or pump); blood glucose monitoring; carbohydrate counting	Lifestyle changes (diet, exercise, weight loss); oral medication (metformin increases cell sensitivity); sometimes insulin
Prevalence	~10% of diabetes cases	~90% of diabetes cases; rapidly increasing worldwide

7.6 Thermoregulation

Humans are **endotherms** — we generate metabolic heat internally and regulate core body temperature at approximately **37 degrees C** (range 36.1–37.8 degrees C). This is optimal for enzyme function. The **hypothalamus** acts as the thermoregulatory centre, functioning as a biological thermostat. It receives information from:

- **Peripheral thermoreceptors** in the skin (detect external temperature changes, providing early warning)
- **Central thermoreceptors** in the hypothalamus itself (detect temperature of blood flowing through it, monitoring core temperature)

Response to Cold (Below Set Point)

Response	Mechanism	Heat Gained By
Vasoconstriction	Smooth muscle in arterioles supplying skin capillaries contracts; blood is diverted to deeper vessels (arteriovenous shunt)	Reduced heat loss by radiation and convection from skin

	vessels). Less warm blood flows near the skin surface	
Shivering	Involuntary, rapid contraction and relaxation of skeletal muscles, controlled by the hypothalamus via motor neurones	Exothermic metabolic reactions in muscle (respiration) generate heat; can increase heat production 5-fold
Piloerection	Erector pili muscles (smooth muscle) contract; body hairs stand upright	Traps a layer of insulating still air next to skin (more effective in furry mammals than humans)
Increased metabolic rate	Hypothalamus stimulates thyroid gland (via anterior pituitary TSH) to release more thyroxine; also stimulates adrenal medulla to release adrenaline	Increased cellular respiration releases more heat as a by-product
Behavioural	Adding clothing, moving to warmer environment, curling up (reducing surface area:volume ratio), huddling	Reduces heat loss and/or gains heat from environment

Response to Heat (Above Set Point)

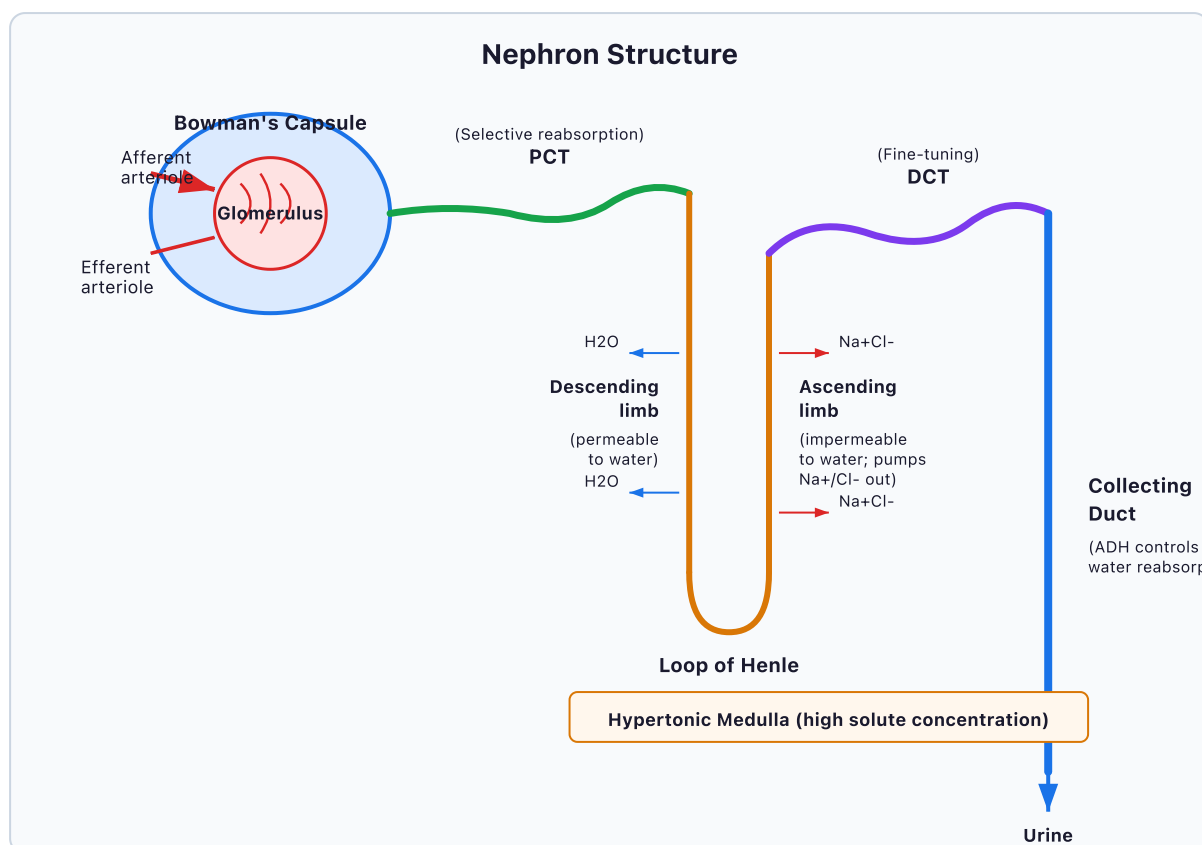
Response	Mechanism	Heat Lost By
Vasodilation	Smooth muscle in arterioles relaxes; more blood flows through superficial capillaries near the skin surface. Skin appears flushed/red	More heat lost by radiation, convection and conduction from skin surface
Sweating	Eccrine sweat glands (innervated by sympathetic cholinergic neurones) secrete sweat (water, NaCl, urea) onto skin surface	Evaporation of sweat absorbs latent heat of vaporisation from the skin (~2,400 kJ per litre evaporated); highly effective cooling mechanism

Hairs lie flat	Erector pili muscles relax; hairs lie against skin	No insulating air layer trapped; heat escapes more easily
Decreased metabolic rate	Reduced thyroxine secretion over longer timescales	Less metabolic heat generated
Behavioural	Removing clothing, moving to shade, spreading out limbs (increasing exposed surface area), fanning, seeking water	Increases heat loss to environment

Exam Tip: A very common error is to say that "blood vessels move closer to the surface" during vasodilation. Capillaries do NOT move. It is the arterioles that dilate (or constrict), controlling the volume of blood flowing through the existing superficial capillary network. Always say "more blood flows through capillaries near the skin surface" rather than "blood vessels move to the surface."

7.7 The Kidney and Osmoregulation

Osmoregulation: The control of the water potential of the blood and body fluids. If blood water potential is too low (too concentrated), cells lose water by osmosis and crenate. If too high (too dilute), cells gain water and may lyse. In mammals, osmoregulation is primarily carried out by the **kidneys** under hormonal control of **ADH** (antidiuretic hormone / vasopressin) from the posterior pituitary gland.



Kidney Functions

The kidneys have three main functions: (1) **excretion** of metabolic waste (urea, creatinine, excess ions), (2) **osmoregulation** (controlling water balance), (3) **regulation of blood pH** and ion concentrations. Each kidney contains ~1 million **nephrons**, the functional units.

Nephron Function (Step by Step)

Region	Process	Detailed Mechanism
Glomerulus + Bowman's capsule	Ultrafiltration	The afferent arteriole is wider than the efferent arteriole, creating high hydrostatic pressure in the glomerular capillaries. This pressure forces small molecules (water, glucose, amino acids, urea, ions, vitamins — all $< \sim 69$ kDa) through the three-layered filter: (1) fenestrated capillary endothelium, (2) basement membrane (glycoprotein mesh acts as molecular sieve), (3) podocyte foot processes (filtration slits). Large molecules (plasma proteins, blood cells, platelets) are too large to pass and remain in the

		blood. The filtrate entering Bowman's capsule is essentially blood plasma minus proteins.
Proximal convoluted tubule (PCT)	Selective reabsorption	~65–85% of filtrate reabsorbed here. Epithelial cells have microvilli (brush border, increasing surface area ~20x), many mitochondria (ATP for active transport), and co-transport proteins . Na^+ is actively pumped out of cells by Na^+/K^+ ATPase on the basal membrane, creating a low Na^+ concentration inside the cell. Na^+ then enters from the filtrate via co-transporter proteins, dragging glucose and amino acids with it (secondary active transport). All glucose and amino acids are reabsorbed (unless blood glucose exceeds renal threshold, ~10 mmol/L, as in uncontrolled diabetes). Water follows by osmosis. Urea is partially reabsorbed passively.
Loop of Henle	Countercurrent multiplier	Descending limb: thin walls, permeable to water but impermeable to ions. Water leaves by osmosis into the increasingly concentrated medullary interstitium. Filtrate becomes progressively more concentrated as it descends. Ascending limb: thick walls (active transport), impermeable to water. $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ co-transporter actively pumps ions out into the medulla (thin ascending = passive, thick ascending = active). Filtrate becomes more dilute as it ascends. The net effect is to create a concentration gradient in the medulla (increasingly hypertonic from cortex to inner medulla), which is essential for water reabsorption in the collecting duct.
Distal convoluted tubule (DCT)	Fine-tuning	Further reabsorption of Na^+ (controlled by aldosterone from adrenal cortex) and Ca^{2+} (controlled by parathyroid hormone). Secretion of H^+ and K^+ to regulate blood pH and potassium levels. Contains the macula densa, part of the juxtaglomerular apparatus that monitors NaCl concentration and regulates renin release.

Collecting duct	ADH-controlled water reabsorption	Passes through the hypertonic medulla. With ADH: ADH binds to V2 receptors on collecting duct cells → cAMP cascade → aquaporin-2 water channels inserted into the apical membrane → water moves out by osmosis into the hypertonic medulla → concentrated urine (small volume). Without ADH: walls remain impermeable to water → dilute urine (large volume). Urea recycling also contributes to the medullary gradient (urea transporters in inner medullary collecting duct).
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ADH and the Osmoregulatory Feedback Loop

Scenario: Dehydration (blood water potential decreases / blood too concentrated)

1. **Osmoreceptors** in the hypothalamus detect decreased blood water potential (these cells lose water by osmosis and shrink/change shape)
2. Hypothalamus stimulates the **posterior pituitary gland** to release more **ADH** into the blood (ADH is actually synthesised in the hypothalamus and transported down neurosecretory neurones to the posterior pituitary for storage)
3. ADH travels in the blood to the kidneys and binds to receptors on **collecting duct** and DCT cells
4. More **aquaporin-2** water channels are inserted into the apical membranes
5. More water is reabsorbed from the collecting duct by osmosis into the hypertonic medulla, then into the vasa recta (capillaries)
6. **Small volume of concentrated (dark) urine** produced
7. Blood water potential returns to normal → osmoreceptors detect this → **less ADH released** (negative feedback)

Scenario: Overhydration (blood water potential increases / blood too dilute)

1. Osmoreceptors detect increased blood water potential (cells gain water by osmosis and swell)
2. Less ADH released from posterior pituitary
3. Fewer aquaporins in collecting duct walls → walls less permeable to water
4. Less water reabsorbed → **large volume of dilute (pale) urine** produced

5. Blood water potential returns to normal (negative feedback)

Exam Tip: Desert animals like the kangaroo rat have **very long loops of Henle** that extend deep into the medulla. This creates a steeper concentration gradient, allowing more water to be reabsorbed from the collecting duct. They can produce urine up to about 14x more concentrated than their blood. Conversely, aquatic mammals (beavers) have short loops and produce dilute urine. This is a common comparative/adaptation question.

7.8 Muscle Contraction

Sliding filament theory: Muscle contraction occurs when **actin (thin) filaments** slide over **myosin (thick) filaments**, shortening the sarcomere. The filaments themselves do not shorten or change length; they increase their degree of overlap. This theory was proposed by Huxley and Hanson in 1954 and is supported by electron microscopy evidence showing changes in banding patterns during contraction.

Structure of Skeletal Muscle (Hierarchical Organisation)

Whole muscle → **fascicles** (bundles wrapped in connective tissue) → **muscle fibres** (individual cells; very large, multinucleate, formed by fusion of myoblasts during development) → **myofibrils** (contractile units within each fibre) → **sarcomeres** (repeating functional units bounded by Z lines).

Key structures:

- **Sarcolemma:** Plasma membrane of the muscle fibre, which extends inward as **T-tubules** (transverse tubules) that carry depolarisation deep into the fibre
- **Sarcoplasmic reticulum (SR):** Specialised smooth ER that stores and releases Ca^{2+} ions; wraps around each myofibril
- **Sarcoplasm:** Cytoplasm of the muscle fibre; rich in glycogen (fuel store) and myoglobin (oxygen store, red pigment)
- **Many mitochondria:** Clustered between myofibrils to supply ATP for contraction

Sarcomere Banding Pattern

Band/Zone	Composition	During Contraction
A band (dark/anisotropic)	Full length of myosin filaments (may overlap with actin at edges)	Stays the SAME length (myosin does not change length)
I band (light/isotropic)	Actin filaments only (no myosin overlap)	Gets SHORTER (more overlap means less actin-only zone)
H zone	Myosin only (centre of A band, no actin overlap)	Gets SHORTER (actin slides in, covering more of the myosin)
Z line (Z disc)	Protein disc anchoring actin filaments; boundary of sarcomere	Z lines move CLOSER together (sarcomere shortens)
M line	Centre of sarcomere; holds myosin filaments in position	Stays in place

Mechanism of Contraction (Sliding Filament + Cross-Bridge Cycle)

1. **Nerve impulse arrives** at the neuromuscular junction (NMJ); ACh released; depolarisation of the sarcolemma
2. Depolarisation spreads along the sarcolemma and down the **T-tubules** into the interior of the muscle fibre
3. The T-tubule depolarisation triggers the **sarcoplasmic reticulum** to release Ca^{2+} ions into the sarcoplasm (via ryanodine receptor channels)
4. Ca^{2+} binds to **troponin** (a regulatory protein on the actin filament); this causes a conformational change in troponin
5. The shape change in troponin moves **tropomyosin** (a filamentous protein wound around actin) away from the myosin-binding sites on the actin filament, exposing them
6. Myosin heads (already energised by previous ATP hydrolysis — ATPase activity of myosin "cocks" the head) bind to the exposed sites on actin, forming **actin-myosin cross-bridges**
7. **Power stroke:** The myosin head pivots (changes angle from ~90 degrees to ~45 degrees), pulling the actin filament towards the centre of the sarcomere. ADP + P_i are released from the myosin head during this stroke

8. A new **ATP** molecule binds to the myosin head, causing it to **detach** from actin (cross-bridge broken)
9. ATP is **hydrolysed** by myosin ATPase → myosin head re-cocked to its high-energy conformation, ready for the next cycle
10. The cycle repeats as long as Ca^{2+} and ATP are available. Each cycle slides actin ~10 nm. Hundreds of myosin heads work asynchronously to produce smooth contraction

Relaxation

When nerve stimulation stops:

1. Ca^{2+} is actively pumped back into the sarcoplasmic reticulum by **Ca^{2+} -ATPase** (SERCA pumps) — this requires ATP
2. Troponin returns to its original conformation; tropomyosin re-covers the myosin-binding sites on actin
3. No further cross-bridges can form; existing ones complete their cycle and detach
4. The muscle returns to its resting length passively (muscle fibres can only pull, not push; an **antagonistic muscle** actively contracts to extend the relaxed muscle)

Exam Tip: During contraction: I bands and H zones get **SHORTER** (more overlap); A bands stay the **SAME** (myosin length unchanged); sarcomere gets shorter (Z lines move closer). This is the most commonly tested aspect of the sliding filament theory. When interpreting electron micrographs: darker regions = more protein overlap; the A band boundary never changes. Also note that **ATP is needed for both contraction (power stroke energy) and relaxation (Ca^{2+} reuptake and cross-bridge detachment)**. This explains rigor mortis: after death, no ATP is produced, so cross-bridges cannot detach and muscles remain rigid.

Muscle Energy Systems

System	Speed	Duration	Substrate
Creatine phosphate (CP)	Immediate (1–2 s)	~10 s	$\text{CP} + \text{ADP} \rightarrow \text{ATP} + \text{creatine}$ (catalysed by creatine kinase)

Anaerobic glycolysis	Fast	~60–90 s	Glucose → pyruvate → lactate (yields 2 ATP per glucose, no O ₂ needed)
Aerobic respiration	Slower to activate	Hours	Glucose/fatty acids + O ₂ → CO ₂ + H ₂ O (yields ~30–32 ATP per glucose)

Slow-Twitch vs Fast-Twitch Muscle Fibres

Feature	Slow-Twitch (Type I)	Fast-Twitch (Type II)
Contraction speed	Slow	Fast
Fatigue resistance	High (resistant to fatigue)	Low (fatigue quickly)
Main energy pathway	Aerobic respiration	Anaerobic glycolysis
Mitochondria	Many	Fewer
Myoglobin content	High (appear red)	Low (appear white/pale)
Blood supply	Rich capillary network	Less extensive
Glycogen stores	Lower	Higher
Suited for	Endurance activities (marathon running, posture maintenance)	Explosive activities (sprinting, jumping, weightlifting)